IEEE Tunisia Section Chapter

Ground Truth Under Attack: Securing the Earth Observation Chain

TSYP14 TECHNICAL CHALLENGE

SCOPE & TOPIC

Earth-observation products support decisions in agriculture, insurance, water management and disaster response.

Their trustworthiness depends on evidence of origin, time, integrity and processing history across the sensing, communication and product chain.

Teams select one segment ground link, in-situ sensing, or imagery and build a verifiable defensive prototype combining the three chapters' expertise:

- Remote sensing (GRSS)
- Signal/image processing (SPS)
- Computational intelligence and security analytics (CIS)

Challenge problem statement

Documented navigation interference, false-data injection and increasingly capable image synthesis show that plausible-looking data may still be untrustworthy.

A failure may remain silent: the pipeline continues to operate while a corrupted observation reaches the analyst.

The challenge focuses on:

- Provenance
- Integrity verification
- Anomaly detection
- Controlled adversarial testing

Challenge Goal

Select one track and defend it:

- Use authentic public data.
- Apply the common standard scenario.
- Detect or prevent the attack.
- Report detection quality, false alarms, latency, memory and security overhead.
- No purchase is required a software or emulated trust anchor and local laptop execution are valid.

Challenge Instructions

Teams choose exactly one track:

Track 1: Secure Ground-Link Testbed

Track 2: False-Data Injection

Track 3: Trust the Pixel

Teams **must** complete six deliverables:

1. Threat model & segment analysis
2. Platform / architecture / trust anchor
3. Signal / image processing
4. AI or statistical detection
5. Adversarial validation & response
6. Documentation using the official reference datasets

Phase 1: **Anonymous** report + code submission

Deadline: **8 November 2026**

Up to 7 finalists advance to ***Phase 2***.

Phase 1 deliverables

- Anonymous Report
- GitHub Repository
- Prototype
- Documentation

The administrative committee assigns a team code and sanitizes the repository before forwarding it to the jury.

Important note:

Phase 1 submissions are fully anonymous, no names, unit names, logos or identifying repository metadata.

Phase 2

Deadline: 11 December 2026

Finalists will build a working prototype and present a live or replayable attack demonstration during TSYP14.

Phase 2 deliverables are non-anonymous due to the live pitch

Rules & Criteria

This challenge is for Student Branches (SB) only.

- The challenge is hosted collaboratively by 3 IEEE chapters.
- Each team must consist of 5 members maximum.
- Teams must choose exactly one track.
- Teams must use the official reference datasets for the standard evaluation scenario.
- Phase 1 submissions must remain anonymous.
- Phase 2 submissions are non-anonymous due to the live pitch.
- Up to 7 teams will be selected as finalists.

Pitching

English | 7 min pitching | 3 min Q&A

The final session will accommodate up to 7 finalist teams within a 1h30 session.

Scoring

TOTAL SCORE: 100 POINTS

Final Score = 30% × Phase 1 Score + 70% × Phase 2 Score

Phase 1: Anonymous Selection: 100 points

Weighted 30% of the final score

- Threat model & relevance: 15 points
- Architecture & trust mechanism: 15 points
- Signal / image processing: 15 points
- Detector design & evaluation: 20 points
- Standard scenario & evasion: 20 points
- Feasibility, affordability & local execution: 10 points
- Clarity & reproducibility: 5 points

Phase 2 — Final Prototype & Pitch: 100 points

Weighted 70% of the final score

- End-to-end functional prototype: 20 points
- Provenance / security control: 15 points
- Processing & local detection: 15 points
- Held-out standard attack & evasion: 20 points
- Metrics, efficiency & accessibility: 10 points
- Documentation & reproducibility: 10 points
- Pitch & technical questions: 10 points

Tie-break order

1. Held-out attack/evasion score
2. Lower false-alarm rate
3. Documented jury decision

Bonus Points

No bonus points are awarded.

Membership/status in GRSS, SPS, CIS, YP or WIE does not affect the score. Hardware, prototype presentation and business-scope coverage are already evaluated within the criteria above.

Winners

Number of winning teams: 3 teams

Prizes:

- **1st Prize:** 800 DT
- **2nd Prize:** 500 DT
- **3rd Prize:** 200 DT

Total Prize Pool: 1500 DT

Important Dates

- Challenge Infosession: **20/09/2026**
- First Submission Deadline: **03/11/2026**
- Preselection Announcement: **08/11/2026**
- Final Deadline: **11/12/2026**

Submission

Submit your project and show us what you've built.

Contact



Please contact us through the following email address ghassen.ltaif@ieee.org for any inquires or clarifications.